Paper 77 — Muon and Its Many Variants

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An exploration of non-Euclidean gradient descent

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The paper in one minute

Question: Which geometry defines the steepest update for a whole neural network, including Muon-trained matrices and Adam-trained parameters?
Model/Method: Choose a norm per layer, combine those norms into one product-space norm, and derive constrained or regularized steepest-descent steps. Test Muon variants with model-truncation momentum (Momo).
Finding: The framework unifies practical Muon+Adam updates and motivates MuonMax, whose Momo version is substantially less sensitive to learning-rate choice in the tested GPT-2 tasks.
Why we care: Optimizer design involves a choice of geometry, not only a learning rate. Better tuning robustness is an empirical finding, not a theorem guaranteeing saddle escape.

1. First principles: "steepest" depends on the norm

For a small update Δ, the first-order model is

$$f(W+\Delta) \approx f(W) + \langle G, \Delta \rangle.$$

Under an Euclidean constraint $\|\Delta\|_F \le \eta$, the steepest update is proportional to $-G$. But a matrix has many legitimate norms. Under a *spectral-norm* constraint, the direction is related to the matrix's polar factor rather than its entrywise values.

$$\Delta_{\mathrm{CSD}} = \arg\min_{\|\Delta\| \le \eta} \langle G, \Delta \rangle, \qquad \Delta_{\mathrm{RSD}} = \arg\min_{\Delta} \left\{ \langle G, \Delta \rangle + \tfrac{1}{2\eta}\|\Delta\|^2 \right\}.$$

These are constrained and regularized steepest descent (CSD and RSD). Their directions and step lengths need not coincide.

2. Worked matrix example: why Muon is not just SGD

Take the illustrative gradient $G = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ and a unit trust-region radius. With Frobenius norm, the steepest unit update is

$$\Delta_F = -\frac{1}{\sqrt{10}} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \langle G, \Delta_F \rangle = -\sqrt{10}.$$

With spectral norm $\|\Delta\|_{2\to 2} \le 1$, its linear minimizer is $\Delta_{\mathrm{sp}} = -I$, yielding

$$\langle G, \Delta_{\mathrm{sp}} \rangle = -4.$$

The spectral trust region permits full unit steps in both singular directions. More generally, if $G = U\Sigma V^\top$, the minimizing matrix is $-UV^\top$ (up to zero singular directions); the dual of spectral norm is the *nuclear norm*, the sum of singular values.

One norm, a different algorithm
A matrix-orthogonalized update can arise from optimizing the same linearized loss under a different trust-region geometry. It is not generally a Newton or inverse-Hessian update.

3. Combine the layers, not just the individual matrices

A network has hidden matrices $W^1, \dots, W^L$ and other parameters θ. The paper uses a norm for each component and aggregates them, for example:

$$\|W\|_\infty = \max_i \|W^i\|_{(i)}, \qquad \|W\|_2 = \left(\sum_i \|W^i\|_{(i)}^2 \right)^{1/2}.$$

Proposition 3.6 interprets practical Muon+Adam as constrained steepest descent for a suitable *product norm* with a spectral norm on hidden matrices and an adaptive norm on other parameters.

4. From Muon to MuonMax

Choosing a different product norm changes how layerwise updates are scaled relative to one another. MuonMax arises as a regularized steepest-descent method under a hybrid norm. Its matrix update includes a scale tied to the *sum of nuclear norms of the layerwise momenta*:

$$W_{t+1}^\ell = W_t^\ell - \eta \left(\sum_j \|M_t^j\|_{\text{nuc}} \right) \text{polar}(M_t^\ell) \quad \text{(schematic matrix part).}$$

In the full method, learning rates and non-matrix parameter rules must also be respected. The layer coupling is important: unlike normalizing each matrix independently, the update scale can reflect the collection of layers.

Why "stale" nuclear norms?

Computing polar factors and their nuclear norms for every layer before updating anything can create memory or runtime overhead. The authors reuse previous-iteration dual norms as an approximation. In their ablations, this reduces the overhead of MuonMax–Momo to about 5% per step relative to MuonAdam, with little effect on measured loss.

5. Model-based momentum: the Momo ingredient

A plain Taylor model can underestimate loss if a chosen step is overly aggressive. Momo adds a known or estimated lower bound F_* for the loss to a truncated first-order model, combining current and historical gradient/loss information. In the paper, integrating Momo with Muon variants improves robustness to step-size tuning.

Do not confuse two questions
The paper *derives a geometric interpretation* of existing and novel optimizers; its preference for MuonMax–Momo comes from language-model experiments. It does not establish a general nonconvex global-convergence or second-order saddle-escape theorem for Muon.

6. Experimental evidence and caveats

The authors sweep 36 optimizer variants on GPT-2 Small trained on FineWeb (1B tokens) and evaluate selected methods on GPT-2 Large with SlimPajama (1B and 6B tokens). Figure 1 and Section 5 report that MuonAdam and Scion can be sensitive to the learning rate, while the Momo variants retain competitive validation loss across a substantially wider range. These are task- and tuning-protocol-dependent results. **Contributions.** (i) A product-norm steepest-descent framework; (ii) an exact geometric interpretation of practical MuonAdam; (iii) new MuonMax updates and efficient stale-norm approximation; (iv) Momo combinations and controlled comparisons.

7. Assumptions and limitations

- Norm equivalences and first-order subproblem derivations assume well-defined linear minimization oracles, dual norms, and the relevant momentum/preconditioning quantities.
- Empirical conclusions come from the selected GPT-2/data/token budgets; they do not prove universal superiority or generalization benefits across architectures.
- The method does not directly solve for Hessian negative-curvature directions. Calling it a guaranteed saddle-escape algorithm would be inaccurate.

8. Quick reference

Euclidean steepest step	$\Delta \propto -G$ under a Frobenius-norm constraint.
Spectral step	$\Delta \propto -\operatorname{polar}(G)$ for matrix weights.
Norm duality	$\|G\|_{\text{nuc}}$ is dual to spectral norm.
CSD vs RSD	Fixed norm-ball radius versus quadratic norm penalty.
MuonAdam	CSD under a mixed spectral/adaptive product norm.
MuonMax	RSD with layer-coupled dual-norm scaling.
Momo	Model truncation uses a loss lower bound.

Papers Master note

Crawshaw et al. interpret Muon+Adam as non-Euclidean steepest descent on network product spaces and introduce MuonMax; Momo improves LR robustness in GPT-2 experiments. Geometric optimizer design and empirical stability are the contributions, not a general saddle-escape theorem.

Original: https://proceedings.mlr.press/v306/crawshaw26a.html. The 2×2 matrix is a pedagogical illustration.